🐍 PYTHON QUICK REVISION NOTES

Beginner → Advanced | Interview + Exam + Real Projects

> **Revision Rule:** Don't memorize Python as isolated syntax.
> Remember **WHY → HOW → WHEN → COMMON MISTAKE → PROJECT USE**.

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0. PYTHON MENTAL MODEL

Python is mainly about:

```
Data → Variables → Logic → Functions → Objects
                      ↓
              Files / Database
                      ↓
              Modules / Packages
                      ↓
              APIs / Backend
                      ↓
                 AI Apps
```

Core question to ask when coding

> **What data do I have → what should happen to it → where should the result go?**

Example:

```
Subtitle text
     ↓
Clean text
     ↓
Extract vocabulary
     ↓
Check existing words
     ↓
Save new words
     ↓
AI meaning / explanation
```

---

1. PYTHON BASICS

1.1 Variables

```
name = "Khushi"
age = 21
score = 87.5
```

Python variables are **references to objects**, not traditional fixed-type boxes.

```
x = 10
x = "hello"
```

Valid because the name `x` can reference different object types.

Remember

```
Variable → reference → object → value/type
```

Common mistake

```
x = 10
y = x
```

`y` does not become a completely independent copy of everything automatically.

---

2. DATA TYPES

Main built-in types

Type	Example	Mutable?
`int`	`10`	No
`float`	`10.5`	No
`str`	`"hello"`	No
`bool`	`True`	No

Type	Example	Mutable?
`list`	`[1, 2, 3]`	Yes
`tuple`	`(1, 2, 3)`	No
`set`	`{1, 2, 3}`	Yes
`dict`	`{"name": "A"}`	Yes
`None`	`None`	No

Interview trap

Mutable ≠ changeable variable.

```
numbers = [1, 2]
numbers.append(3)
```

The list object changed.

```
x = 10
x += 1
```

An integer is immutable; Python creates/references another integer object.

---

3. STRINGS

```
word = "Python"
```

Useful operations:

```
word.lower()
word.upper()
word.strip()
word.replace("Py", "My")
word.split()
"-".join(["AI", "Learning"])
```

Important distinction

```
split()  → string → list
join()   → iterable of strings → string
```

Example:

```
sentence = "learn python fast"

words = sentence.split()

result = "-".join(words)
```

Hidden mistake

```
"hello".replace("h", "H")
```

doesn't modify the original string.

Strings are immutable.

You need:

```
text = text.replace("h", "H")
```

---

4. OPERATORS

Arithmetic

```
+  -  *  /  //  %  **
```

Important

```
10 / 3   # 3.333...
10 // 3  # 3
10 % 3   # 1
```

Comparison

```
==   !=   >   <   >=   <=
```

Critical interview distinction

```
==   # value equality
is   # object identity
```

Usually:

```
x == y
```

for comparing values.

Don't casually use:

```
x is y
```

---

5. BOOLEAN LOGIC

```
and
or
not
```

Example:

```
if age >= 18 and has_id:
    print("Allowed")
```

Short-circuit evaluation

Python may stop evaluating early.

```
if user and user.is_active:
    ...
```

If `user` is falsy, Python doesn't evaluate the second condition.

Hidden mistake

Know the difference between:

```
if value:
```

and:

```
if value is not None:
```

These are NOT equivalent.

`0`, `""`, `[]`, `{}` are falsy too.

---

6. CONDITIONAL STATEMENTS

```
if condition:
    ...
elif condition:
    ...
else:
    ...
```

Practical pattern

```
if task["status"] == "completed":
    ...
elif task["priority"] == "high":
    ...
else:
    ...
```

Project connection

To-Do App:

```
Pending
Completed
Overdue
High Priority
```

---

7. LOOPS

for loop

```
for word in words:
    print(word)
```

while loop

```
while condition:
    ...
```

Use

```
for    → iterate over known collection/range
while  → continue while condition remains true
```

Hidden mistakes

Infinite loop:

```
while condition:
    ...
```

without changing what eventually makes `condition` false.

---

8. RANGE

```
range(start, stop, step)
```

Important:

- `stop` is excluded.

```
range(1, 5)
```

produces:

```
1 2 3 4
```

---

9. LISTS

```
words = ["hello", "python", "anime"]
```

Useful:

```
append()
extend()
insert()
remove()
pop()
clear()
sort()
reverse()
```

Important difference

```
append([3, 4])
```

adds ONE item.

```
extend([3, 4])
```

adds multiple items.

---

10. LIST INDEXING & SLICING

```
items[0]
items[-1]
items[1:4]
items[:3]
items[::2]
```

Hidden mistake

```
items[1:4]
```

means indexes:

```
1, 2, 3
```

NOT 4.

---

11. TUPLES

```
point = (10, 20)
```

Useful when data should not be modified.

```
x, y = point
```

Interview idea

Tuple immutability can make it useful for fixed collections and certain dictionary/set operations.

---

12. SETS

```
words = {"python", "java", "python"}
```

Result:

```
{"python", "java"}
```

Useful for:

- removing duplicates
- membership checking
- set operations

```
a | b
a & b
a - b
```

Project connection

Subtitle vocabulary:

```
Extracted words
       ↓
set()
       ↓
remove duplicates
```

---

13. DICTIONARIES

```
student = {
    "name": "Khushi",
    "score": 90
}
```

Access:

```
student["name"]
```

Safer:

```
student.get("name")
```

Important difference

```
student["age"]
```

can raise `KeyError`.

```
student.get("age")
```

returns `None` if missing.

Dictionary iteration

```
for key, value in data.items():
    ...
```

Also:

```
data.keys()
data.values()
data.items()
```

Project connection

AI vocabulary:

```
vocabulary = {
    "beautiful": "attractive",
    "rapid": "very fast"
}
```

---

14. COMPREHENSIONS

List:

```
squares = [x * x for x in numbers]
```

With condition:

```
even = [x for x in numbers if x % 2 == 0]
```

Dictionary:

```
squares = {x: x*x for x in numbers}
```

Interview warning

Comprehensions are useful, but don't sacrifice readability.

Bad:

```
result = [x for x in data if ... if ... if ...]
```

Sometimes a normal loop is clearer.

---

15. FUNCTIONS

```
def clean_text(text):
    return text.strip().lower()
```

Function anatomy

```
def
 ↓
name
 ↓
parameters
 ↓
```

```
body
 ↓
return value
```

`return` vs `print`

```
def add(a, b):
    return a + b
```

`return` sends a value back.

```
print(add(2, 3))
```

`print()` only displays something.

Hidden mistake

```
def add(a, b):
    print(a + b)
```

This function returns:

```
None
```

---

16. PARAMETERS

```
def greet(name):
    ...
```

Default:

```
def greet(name="User"):
    ...
```

Keyword:

```
greet(name="Khushi")
```

Variable arguments:

```
def total(*numbers):
    return sum(numbers)
```

Keyword arguments:

```
def create_user(**data):
    ...
```

---

17. SCOPE

Main concepts:

```
Local
Enclosing
Global
Built-in
```

LEGB:

```
L → Local
E → Enclosing
G → Global
B → Built-in
```

Hidden mistake

Avoid unnecessary global variables.

Prefer:

```
def process(data):
    return result
```

over functions secretly depending on global state.

---

18. EXCEPTIONS

```
try:
    result = 10 / 0
except ZeroDivisionError:
    print("Cannot divide by zero")
```

Can also use:

```
else:
finally:
```

Pattern:

```
try:
    ...
except SpecificError:
    ...
else:
    ...
finally:
    ...
```

Important rule

Catch specific exceptions.

Prefer:

```
except FileNotFoundError:
```

over blindly using:

```
except Exception:
```

---

19. FILE HANDLING

Basic:

```
with open("data.txt", "r") as file:
    content = file.read()
```

Modes:

```
r  → read
w  → write
a  → append
x  → create
b  → binary
```

VERY IMPORTANT

Prefer:

```
with open(...)
```

because the context manager handles closing the file.

---

20. JSON

Very important for your current projects.

```
import json
```

Python → JSON:

```
json.dump(data, file)
```

JSON → Python:

```
data = json.load(file)
```

String conversion:

```
json.dumps()
json.loads()
```

Remember

```
dump  → file
dumps → string

load  → file
loads → string
```

---

21. MODULES

```
import math
```

or:

```
from math import sqrt
```

Best practice

Prefer clear imports.

Avoid:

```
from module import *
```

because it makes dependencies unclear.

---

22. PACKAGES

Typical project:

```
app/
├── main.py
├── services/
├── models/
├── utils/
└── config/
```

Package = organized collection of Python modules.

Project connection

Your To-Do App and AI Subtitle project can be separated into:

```
UI
Business Logic
Data
Utilities
Configuration
```

---

23. OOP

Core concepts:

```
Class
Object
Attribute
Method
Encapsulation
Inheritance
Polymorphism
Abstraction
```

Example:

```python
class Vocabulary:
    def __init__(self, word):
        self.word = word
```

```
    def show(self):
        return self.word
```

Object:

```
item = Vocabulary("beautiful")
```

Remember

```
Class  → blueprint
Object → actual instance
```

---

24. `self`

```
self.word
```

means:

> the `word` attribute belonging to this particular object.

Common mistake:

```
class User:
    def __init__(name):
        ...
```

Correct:

```
class User:
    def __init__(self, name):
        self.name = name
```

---

25. ENCAPSULATION

Keep related data + behavior together.

Instead of:

```
raw data everywhere
functions everywhere
```

use:

```
Object
 ├── data
 └── behavior
```

Project connection

```
class Task:
    title
    category
    priority
    status
```

---

26. INHERITANCE

```
class Animal:
    ...

class Dog(Animal):
    ...
```

Dog receives behavior from Animal.

Practical warning

Don't use inheritance simply because you can.

Often **composition** is cleaner.

---

27. POLYMORPHISM

Different objects can provide the same interface.

```
class PDFReader:
    def read(self):
        ...

class TextReader:
    def read(self):
        ...
```

Code can work with either if it only needs:

```
reader.read()
```

Very useful in scalable applications.

---

28. SPECIAL METHODS

Examples:

```
__init__
__str__
__repr__
__len__
__eq__
```

Example:

```
def __str__(self):
    return self.name
```

Important

`__init__` initializes an already-created object.

It is not technically the object's constructor in the strictest Python terminology.

---

29. ITERATORS & ITERABLES

Iterable:

Something you can iterate over.

Examples:

```
list
tuple
string
dict
set
```

Iterator:

Object that produces values one at a time.

Core function:

```
iter()
next()
```

---

30. GENERATORS

```
def numbers():
    yield 1
    yield 2
    yield 3
```

Generator values are produced lazily.

Why useful?

For large data:

```
Huge file
   ↓
read everything ❌

Huge file
   ↓
process one piece at a time ✅
```

Project connection

Processing a huge subtitle/video dataset.

---

31. LAMBDA

```
square = lambda x: x * x
```

Useful for small functions.

Often used with:

```
sorted()
map()
filter()
```

Don't force lambda everywhere.

A normal function is often more readable.

---

32. `map`, `filter`, `sorted`

```
map()
filter()
sorted()
```

Example:

```
sorted(words, key=len)
```

Important

`sorted()` returns a **new list**.

`list.sort()` modifies the existing list and returns `None`.

---

33. DECORATORS

A decorator modifies/extents function behavior.

Conceptually:

```
Original function
       ↓
Decorator
       ↓
Enhanced function
```

Used heavily in:

- Flask
- FastAPI
- authentication
- logging
- authorization
- routing

Example concept:

```
@app.get("/users")
def users():
    ...
```

---

34. CONTEXT MANAGERS

Used for resource management.

```
with open(...) as file:
    ...
```

Common resources:

```
files
database connections
```

```
locks
network resources
```

---

35. TYPE HINTS

```python
def add(a: int, b: int) -> int:
    return a + b
```

Useful for:

- readability
- IDE support
- static analysis
- larger projects

Python remains dynamically typed.

Type hints do not automatically enforce types at runtime.

---

36. DATACLASSES

Useful for classes mainly representing data.

```python
from dataclasses import dataclass

@dataclass
class Task:
    title: str
    completed: bool = False
```

Can reduce boilerplate.

Useful for structured application data.

---

37. TESTING

Basic:

```
assert add(2, 3) == 5
```

Professional projects should use testing frameworks such as:

```
pytest
unittest
```

Test:

```
normal input
edge cases
invalid input
expected failures
```

Hidden mistake

Testing only the happy path is not enough.

---

38. LOGGING

Avoid using `print()` as your application's permanent logging system.

Use:

```
import logging
```

Typical levels:

```
DEBUG
INFO
WARNING
ERROR
CRITICAL
```

Useful for debugging real applications.

---

39. DATABASE BASICS

Python applications commonly interact with:

```
SQLite
PostgreSQL
MySQL
```

Important concepts:

```
Table
Row
Column
Primary Key
Foreign Key
CRUD
Transaction
Query
```

CRUD:

```
Create
Read
Update
Delete
```

Project connection

Your To-Do App:

```
Task
 ├── id
 ├── title
 ├── category
 ├── priority
 ├── status
 └── created_at
```

---

40. SQL + PYTHON

Typical architecture:

```
UI
 ↓
Service / Business Logic
 ↓
Database Layer
 ↓
SQL Database
```

Avoid putting SQL everywhere inside UI code.

---

41. APIS

API = interface through which applications communicate.

Typical HTTP methods:

```
GET     → retrieve
POST    → create
PUT     → replace/update
PATCH   → partial update
DELETE  → delete
```

Common status codes:

```
200 → success
201 → created
400 → bad request
401 → unauthenticated
403 → forbidden
404 → not found
500 → server error
```

---

42. BACKEND ARCHITECTURE

Think:

```
Client
  ↓
API Route
  ↓
Validation
  ↓
Business Logic
  ↓
Database
```

Professional application:

```
Presentation
     ↓
Controller / Route
     ↓
Service
     ↓
Repository / Data Access
     ↓
Database
```

---

43. VIRTUAL ENVIRONMENTS

Create isolated project environments.

```
Project A
 └── dependencies

Project B
 └── different dependencies
```

Avoid installing every package globally.

Typical:

```
python -m venv .venv
```

---

44. DEPENDENCY MANAGEMENT

Know:

```
pip
requirements.txt
pyproject.toml
virtual environment
```

Example:

```
pip install requests
```

---

45. PATH & OS

Very important for your File Organizer.

Use:

```
from pathlib import Path
```

Prefer `pathlib` for modern path handling.

Example concept:

```
Path("downloads") / "file.txt"
```

Why?

Cleaner and more cross-platform than manually building:

```
"C:\\folder\\file.txt"
```

46. AUTOMATION

Your Smart File Organizer uses:

```
Path
file extensions
directories
file operations
error handling
logging
```

Pipeline:

```
Scan folder
 ↓
Identify files
 ↓
Classify
 ↓
Create folders
 ↓
Move files
 ↓
Log operation
 ↓
Handle errors
```

47. REGULAR EXPRESSIONS

Module:

```
import re
```

Useful for pattern matching.

Examples:

```
email
phone number
subtitle timestamps
special patterns
```

But don't use regex when simple string operations are enough.

---

48. MEMORY & PERFORMANCE

Important concepts:

```
Mutable vs immutable
references
copy vs deep copy
lazy evaluation
generators
algorithm complexity
```

Shallow copy trap

```
a = [[1, 2], [3, 4]]
b = a.copy()
```

The outer list is copied, but nested lists are still shared.

For deeply nested independent data:

```
import copy
b = copy.deepcopy(a)
```

---

49. TIME COMPLEXITY

Basic awareness:

```
O(1)      constant
O(log n)  logarithmic
O(n)      linear
```

```
O(n log n)
O(n²)     quadratic
```

Example:

```
if item in my_list:
```

generally requires linear search.

For frequent membership checks:

```
if item in my_set:
```

average-case membership is much faster.

Project connection

Subtitle vocabulary lookup:

```
List search repeatedly → slower
Set/dict lookup → usually better
```

---

50. CONCURRENCY BASICS

Know the concepts:

```
Threading
Multiprocessing
Asyncio
```

Very simplified:

```
Threading
→ useful for many I/O-bound tasks

Multiprocessing
→ useful for CPU-heavy work
```

```
Asyncio
→ useful for cooperative asynchronous I/O
```

Do not blindly use threads for CPU-heavy Python code.

---

51. ASYNC / AWAIT

```python
async def fetch_data():
    ...
```

and:

```python
await fetch_data()
```

Useful when applications perform many waiting operations such as network requests.

Common backend use:

```
API calls
database/network I/O
multiple external services
```

---

52. API + AI INTEGRATION

Your AI project can follow:

```
User Input
    ↓
Validation
    ↓
Prompt / Request Preparation
    ↓
AI API
    ↓
Response Validation
    ↓
Application Logic
```

```
    ↓
Database / File
```

Hidden mistake

Never blindly trust AI output.

Validate:

```
format
required fields
length
expected values
errors
```

---

53. ENVIRONMENT VARIABLES

Never hard-code secrets.

Bad:

```
API_KEY = "my-secret-key"
```

Better:

```
Environment variable
        ↓
Application
```

Use `.env` where appropriate and never commit secrets to GitHub.

---

54. ERROR HANDLING — PROFESSIONAL LEVEL

Good error handling answers:

```
What failed?
Why did it fail?
Can we recover?
```

```
What should the user see?
What should be logged?
```

Don't do:

```
try:
    ...
except:
    pass
```

This can silently hide serious bugs.

---

55. DEBUGGING MINDSET

When something breaks:

```
1. Reproduce
2. Read the error
3. Identify exact line
4. Check input/state
5. Form hypothesis
6. Test hypothesis
7. Fix
8. Test again
9. Document if important
```

Golden rule

> Don't fix the error message. Find the underlying cause.

---

56. GIT + PYTHON PROJECTS

Know:

```
repository
commit
branch
pull
push
```

```
merge
.gitignore
README
```

Important Python `.gitignore`:

```
__pycache__/
*.pyc
.venv/
.env
```

Hidden mistake

`.gitignore` prevents **new untracked files** from being added.

If a file is already tracked, `.gitignore` alone doesn't remove it from Git tracking.

---

57. PROFESSIONAL PROJECT STRUCTURE

Example:

```
project/
│
├── app/
│   ├── models/
│   ├── services/
│   ├── repositories/
│   ├── utils/
│   └── config/
│
├── tests/
├── docs/
├── .env.example
├── .gitignore
├── README.md
├── requirements.txt
└── main.py
```

Don't blindly copy this structure.

Architecture should match project complexity.

58. PYTHON INTERVIEW "TRAPS"

Trap 1 — Mutable default argument

Avoid:

```
def add_item(item, items=[]):
    items.append(item)
    return items
```

The default list is reused.

Better:

```
def add_item(item, items=None):
    if items is None:
        items = []

    items.append(item)
    return items
```

Trap 2 — `is` vs `==`

```
== → equality
is → identity
```

Trap 3 — `sort()` vs `sorted()`

```
list.sort() → modifies list, returns None

sorted(list) → creates sorted result
```

Trap 4 — `append()` vs `extend()`

```
append(x) → one item
extend(x) → items from iterable
```

Trap 5 — `print()` vs `return`

```
print → display
return → send result back
```

Trap 6 — Mutable vs immutable

```
list/dict/set → mutable

str/int/float/tuple/bool → immutable
```

Trap 7 — Shallow vs deep copy

Nested mutable objects can still be shared after shallow copy.

Trap 8 — Exception swallowing

```
except:
    pass
```

can hide bugs.

Trap 9 — Modifying a collection while iterating

Dangerous pattern:

```
for item in items:
    if condition:
        items.remove(item)
```

Prefer constructing a new collection or iterating over a copy when appropriate.

Trap 10 — `None`

`None` means absence of a value.

It is not:

```
0
False
""
```

---

59. REAL-LIFE PRACTICE PROBLEMS

Problem 1 — Subtitle Vocabulary Cleaner

Connected project: AI Subtitle Learning Platform

Input:

```
"Hello!!! I am learning Python, Python is amazing."
```

Build a function that:

```
1. lowercase text
2. remove punctuation
3. split into words
4. remove duplicates
5. return vocabulary
```

Expected concepts:

```
str
loops
sets
functions
```

Problem 2 — Smart Vocabulary Memory

Input:

```
existing = {
    "beautiful": "attractive",
    "rapid": "very fast"
}
```

New subtitle:

```
"Beautiful scenes move rapidly."
```

Build logic that:

```
1. identifies words
2. checks existing vocabulary
3. detects new words
4. stores them
```

Expected concepts:

```
dict
set
functions
file handling
JSON
```

Problem 3 — Subtitle Sentence Analyzer

Given:

```
"I really love this beautiful scene."
```

Return:

```
word_count
character_count
unique_word_count
longest_word
```

Practice:

```
strings
lists
sets
functions
```

Problem 4 — Smart File Organizer

Connected project: Smart File Organizer

Given:

```
Downloads/
    resume.pdf
    photo.jpg
    song.mp3
    notes.txt
    report.pdf
```

Build a program that classifies:

```
Documents/
Images/
Audio/
Other/
```

Practice:

```
pathlib
file extensions
loops
functions
```

```
exceptions
automation
```

Problem 5 — To-Do Task Manager

Connected project: To-Do App

Represent:

```
task = {
    "id": 1,
    "title": "Study Python",
    "category": "Study",
    "priority": "High",
    "status": "Pending"
}
```

Implement:

```
create task
read tasks
update task
delete task
filter by category
filter by status
```

Practice:

```
dict
list
functions
CRUD
JSON
database preparation
```

Problem 6 — Duplicate Vocabulary Detector

Given 10,000 subtitle words.

Design a solution that quickly determines:

> "Have I already seen this word?"

Compare:

```
list
set
dictionary
```

Explain which one you would choose and why.

Practice:

```
data structures
time complexity
memory
```

Problem 7 — AI Response Validator

Suppose AI returns:

```
{
    "word": "beautiful",
    "meaning": "attractive",
    "example": "She has a beautiful voice."
}
```

Write validation that checks:

```
word exists
meaning exists
example exists
values are strings
```

Practice:

```
dict
functions
```

```
exceptions
validation
AI integration
```

Problem 8 — File Organizer Safety System

Before moving a file:

```
1. Does source exist?
2. Is destination available?
3. Does destination filename already exist?
4. Is the file actually a file?
5. Can operation fail due to permission?
```

Build safe logic.

Practice:

```
pathlib
exceptions
defensive programming
```

Problem 9 — To-Do Analytics

Given 100 tasks, calculate:

```
total tasks
completed tasks
pending tasks
high-priority tasks
completion percentage
most-used category
```

Practice:

```
loops
dict
```

```
functions
statistics
data processing
```

Problem 10 — Mini Backend Challenge

Design an API for your To-Do App.

Endpoints:

```
GET    /tasks
POST   /tasks
GET    /tasks/{id}
PUT    /tasks/{id}
DELETE /tasks/{id}
```

For each endpoint define:

```
input
validation
business logic
database operation
response
error cases
```

This tests whether you can move from:

```
Python knowledge
       ↓
Application design
       ↓
Backend development
```

---

60. YOUR PYTHON LEARNING LADDER

LEVEL 1 — FOUNDATION

Master:

```
variables
data types
operators
strings
conditions
loops
lists
tuples
sets
dictionaries
functions
```

Goal:

> Build small programs without copying logic blindly.

LEVEL 2 — PRACTICAL PYTHON

Master:

```
files
JSON
exceptions
modules
packages
comprehensions
OOP
pathlib
logging
testing
```

Goal:

> Build maintainable small applications.

LEVEL 3 — PROFESSIONAL PYTHON

Master:

```
OOP design
type hints
```

```
dataclasses
iterators
generators
decorators
context managers
testing
logging
architecture
Git
virtual environments
dependencies
```

Goal:

> Write code another developer can understand and maintain.

LEVEL 4 — BACKEND PYTHON

Master:

```
HTTP
REST APIs
FastAPI/Flask
request/response
validation
authentication
database
SQL
ORM
API errors
async/await
deployment basics
```

Goal:

> Build real backend applications.

LEVEL 5 — AI APPLICATION PYTHON

Master:

```
API integration
prompt construction
structured outputs
validation
embeddings
vector databases
RAG concepts
async AI calls
cost/error handling
security
evaluation
```

Goal:

> Build AI-powered applications rather than merely calling an AI API.

---

61. INTERVIEW REVISION FORMULA

For every Python topic, remember these 5 questions:

```
1. WHAT is it?
2. WHY do we use it?
3. WHEN should I use it?
4. WHAT commonly goes wrong?
5. WHERE have I used it in my projects?
```

Example — `set`

```
WHAT?
Collection of unique elements.

WHY?
Fast membership checking and duplicate removal.

WHEN?
When uniqueness/membership matters.

COMMON MISTAKE?
Assuming it preserves list-like indexing/order semantics.
```

```
PROJECT?
Subtitle vocabulary deduplication.
```

---

62. EXAM REVISION FORMULA

For theory questions:

```
Definition
     ↓
Syntax
     ↓
Example
     ↓
Advantages
     ↓
Limitations
     ↓
Practical use
```

For programming questions:

```
Understand input
     ↓
Identify data structure
     ↓
Write logic
     ↓
Handle edge cases
     ↓
Test
```

---

63. YOUR MOST IMPORTANT PYTHON CHECKLIST

Before calling yourself interview-ready, you should be comfortable explaining:

```
□ Mutable vs immutable
□ List vs tuple
□ List vs set
```

```
□ List vs dictionary
□ == vs is
□ append vs extend
□ sort vs sorted
□ print vs return
□ shallow vs deep copy
□ local/global scope
□ LEGB
□ *args / **kwargs
□ exception handling
□ with/context manager
□ iterator vs iterable
□ generator
□ decorator
□ OOP principles
□ self
□ inheritance vs composition
□ type hints
□ virtual environment
□ JSON
□ pathlib
□ logging
□ testing
□ CRUD
□ HTTP methods
□ API status codes
□ SQL basics
□ async/await
□ time complexity
□ Git/.gitignore
```

---

64. FINAL PROJECT-BASED TEST

Don't finish Python by only solving:

```
factorial
prime number
palindrome
calculator
```

Those are useful for fundamentals, but your **real test** should be:

Project A — AI Subtitle Learning Platform

Build:

```
SRT file
 ↓
Parser
 ↓
Text cleaner
 ↓
Vocabulary extractor
 ↓
Duplicate detector
 ↓
Vocabulary memory
 ↓
JSON/database
 ↓
AI explanation
 ↓
Learning UI
```

Project B — Smart File Organizer

Build:

```
Folder
 ↓
Scanner
 ↓
File classifier
 ↓
Duplicate/safety checks
 ↓
Organizer
 ↓
Logging
 ↓
Error recovery
```

Project C — To-Do Application

Build:

```
UI
 ↓
Task service
 ↓
Validation
 ↓
CRUD
 ↓
Database
 ↓
Filtering
 ↓
Analytics
```

Project D — AI/API Backend

Build:

```
Client
 ↓
API
 ↓
Validation
 ↓
Service layer
 ↓
Database
 ↓
AI service
 ↓
Response
```

If you can build and explain these four systems, your Python knowledge becomes **practical portfolio knowledge**, not just interview memorization.

⚡ 30-SECOND PYTHON REVISION

```
DATA
↓
Choose the right data structure
↓
LOGIC
```

```
↓
Functions keep logic reusable
↓
OOP organizes complex systems
↓
FILES / JSON / DATABASE
↓
Persist information
↓
ERROR HANDLING
↓
Make application reliable
↓
TESTING + LOGGING
↓
Make it maintainable
↓
API + BACKEND
↓
Connect applications
↓
ASYNC + PERFORMANCE
↓
Scale I/O-heavy systems
↓
AI INTEGRATION
↓
Build intelligent applications
```

The real goal is not "I know Python syntax."

The goal is: "Give me a problem, and I can choose the right data structure, design the logic, handle failure, test it, and turn it into a maintainable application."